

Inverse of a Matrix

Definition 8.2.1 An $n \times n$ matrix A is called **invertible** (or **nonsingular**) if there exists an $n \times n$ matrix B such that

$$AB = BA = I_n. \quad (1)$$

Here I_n is the identity matrix of order n which has 1's in the main diagonal and 0's everywhere else, namely

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

If the order of I_n is clear from the context, we will simply write I for I_n .

Here B is called **inverse** of A . Since the eq. (1) is symmetric in A and B , we can also say A is an inverse of B .

For example, $A = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$ is invertible since there exists a matrix $B = \begin{bmatrix} -1 & 4 & -2 \\ 2 & -7 & 4 \\ -1 & 2 & -1 \end{bmatrix}$ such that

$$AB = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 4 & -2 \\ 2 & -7 & 4 \\ -1 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I,$$
$$BA = \begin{bmatrix} -1 & 4 & -2 \\ 2 & -7 & 4 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

Not every square matrix is invertible. If A is not an invertible matrix, then we say A is **noninvertible** (or **singular**).

For example, the zero matrix $O = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is noninvertible. Clearly, there cannot exist any matrix B such that $OB = BO = I$ since $OB = BO = O$.

Inverse of a Matrix

Theorem (Uniqueness of an Inverse)

If A is an invertible matrix, then its inverse is unique.

Proof. Let A be an $n \times n$ invertible matrix. So, there exists an $n \times n$ matrix B such that

$$AB = BA = I.$$

Let us assume that there exists another $n \times n$ matrix C such that

$$AC = CA = I.$$

To show that $B = C$, we use

$$\begin{aligned} B &= BI && \text{[Definition of Identity matrix]} \\ &= B(AC) && \text{[} C \text{ is an inverse of } A \text{]} \\ &= (BA)C && \text{[Associative Property for Matrix Multiplication]} \\ &= IC && \text{[} B \text{ is an inverse of } A \text{]} \\ &= C && \text{[Definition of Identity matrix]} \end{aligned}$$

The proof is complete. ■

Notation: If A is invertible and since its inverse is unique, we will use the symbol A^{-1} to denote *the* inverse of A .

Properties of Inverse of a Matrix

Theorem If A be an $n \times n$ invertible matrix, then the following are true.

1. $(A^{-1})^{-1} = A$.
2. $(A^k)^{-1} = \underbrace{A^{-1}A^{-1} \dots A^{-1}}_{k \text{ many factors}} = (A^{-1})^k$
3. $(cA)^{-1} = c^{-1}A^{-1}$ if $c \neq 0$.
4. $(A^T)^{-1} = (A^{-1})^T$

Proof.

1. By definition,

$$A^{-1}(A^{-1})^{-1} = I$$

Now by multiplying both sides by A from the left, we get the result as follows

$$A(A^{-1}(A^{-1})^{-1}) = AI$$

$$(AA^{-1})(A^{-1})^{-1} = A$$

$$I(A^{-1})^{-1} = A$$

$$(A^{-1})^{-1} = A$$

2. By definition,

$$A^k(A^k)^{-1} = I$$

Since k is a positive integer,

$$\underbrace{(AA \dots A)}_k (A^k)^{-1} = I$$

Now multiplying both sides by A^{-1} from the left by k many times we obtain the desired result.

$$(A^k)^{-1} = \underbrace{(A^{-1}A^{-1} \dots A^{-1})}_k = (A^{-1})^k$$

3. By definition,

$$(cA)(cA)^{-1} = I$$

$$A(cA)^{-1} = c^{-1}I$$

Properties of Inverse of a Matrix

$$(cA)^{-1} = A^{-1}(c^{-1}I) = c^{-1}(A^{-1}I) = c^{-1}A^{-1}$$

4. By definition,

$$AA^{-1} = I$$

Taking the transpose of both sides, we obtain

$$(AA^{-1})^T = I^T$$

$$(A^{-1})^T A^T = I$$

Here we have used $(AB)^T = B^T A^T$ and $I^T = I$.

$$(A^{-1})^T A^T = I$$

$$\therefore (A^T)^{-1} = (A^{-1})^T.$$

Theorem. If A and B are two $n \times n$ invertible matrices, then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

(This is known as **Socks-shoes property**. You might be surprised to see that taking the multiplicative inverse reverses the order of multiplication. So interpret A as putting on socks, and B as putting on shoes. To reverse the operation AB of putting on both socks and shoes, you must reverse the order: you take off shoes first, then the socks, and so the inverse operation is $B^{-1}A^{-1}$.)

Proof. The proof is straightforward. Observe that

$$\begin{aligned}(B^{-1}A^{-1})(AB) &= (B^{-1}(A^{-1}A))B \\ &= (B^{-1}I)B \\ &= B^{-1}B \\ &= I\end{aligned}$$

Properties of Inverse of a Matrix

Similarly,

$$\begin{aligned}(AB)(B^{-1}A^{-1}) &= (A(BB^{-1}))A^{-1} \\ &= (AI)A^{-1} \\ &= AA^{-1} \\ &= I\end{aligned}$$

Therefore, AB is invertible and $B^{-1}A^{-1}$ is an inverse of AB . Since inverse of an invertible matrix is unique, $B^{-1}A^{-1}$ is the inverse of AB . Thus

$$(AB)^{-1} = B^{-1}A^{-1}. \blacksquare$$

Corollary: If $A_1, A_2, \dots, A_n$ are $n \times n$ invertible matrices, then

$$(A_1A_2 \cdots A_n)^{-1} = A_n^{-1} \cdots A_2^{-1}A_1^{-1}.$$

In general, $AC = BC$ does not imply $A = B$. In other words, cancellation property does not hold for matrices. For example, if

$$A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix}$$

then $AC = \begin{bmatrix} -2 & 4 \\ -1 & 2 \end{bmatrix} = BC$. However, $A \neq B$.

The next theorem tells us under what condition matrix equation enjoys the cancellation property.

Properties of Inverse of a Matrix

Theorem Let A, B , and C be $n \times n$ matrices. If C is invertible, then

1. If $AC = BC$, then $A = B$. This is called the **right cancellation property**.
2. If $CA = CB$, then $A = B$. This is called the **left cancellation property**.

Proof.

1. Since C^{-1} exists, we have

$$A = AI = A(CC^{-1}) = (AC)C^{-1} = (BC)C^{-1} = B(CC^{-1}) = BI = B$$

The proof of 2. is similar. ■

Caution that $AC = CB$ does not imply $A = B$ even if C is invertible unless $AC = CA$.

Properties of Inverse of a Matrix

Theorem. If A is invertible, then the system $A\mathbf{x} = \mathbf{b}$ has unique solution $\mathbf{x} = A^{-1}\mathbf{b}$

Proof.

$$\mathbf{x} = I\mathbf{x} = (A^{-1}A)\mathbf{x} = A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{b}.$$

The proof is complete. ■

Definition Two matrices A and B are called **row equivalent** if one can be obtained from the other by a sequence of elementary row operations.

For example, the matrices $A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & -1 \\ 6 & 1 & -4 \\ 1 & 2 & 3 \end{bmatrix}$ are row equivalent since

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} \xrightarrow{\substack{4R_1+R_2 \rightarrow R_2 \\ -2R_1+R_3 \rightarrow R_3}} \begin{bmatrix} 1 & 0 & -1 \\ 6 & 1 & -4 \\ 1 & 2 & 3 \end{bmatrix} = B.$$

Theorem. A is invertible if and only if A is row equivalent to I .

Proof. ($\Rightarrow$) Let A be invertible, then the matrix equation $A\mathbf{x} = \mathbf{b}$ has a unique solution $I\mathbf{x} = \mathbf{x} = A^{-1}\mathbf{b}$. Therefore, by Gauss-Jordan elimination process, A can be reduced to I by a sequence of elementary row operations. Hence, A is row equivalent to I .

Conversely, if A is row equivalent to I , then there exists a sequence of elementary matrices $E_1, E_2, \dots, E_n$ corresponding to each elementary row operations such that

$$E_n \cdots E_2 E_1 A = I$$

Define $B = E_n \cdots E_2 E_1$. Hence A is invertible and B is the inverse of A .

How to find A^{-1} ?

$$\underbrace{(E_n \cdots E_2 E_1)}_{A^{-1}} A = I$$

How to find A^{-1} ?

Therefore, by the uniqueness of inverse, we have

$$A^{-1} = E_n \cdots E_2 E_1 I.$$

Therefore, the formula says that one needs to apply the same sequence of elementary row operations to an identity matrix I that helped A reduces to I .

Example. Find A^{-1} for

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix},$$

if exists.

Solution.

$$\begin{aligned} [A|I] &= \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 & 0 & 1 \end{array} \right] \\ &\xrightarrow{\substack{-2R_1+R_2 \rightarrow R_2 \\ -3R_1+R_3 \rightarrow R_3}} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 4 & -2 & 1 & 0 \\ 0 & 2 & 7 & -3 & 0 & 1 \end{array} \right] \\ &\xrightarrow{-2R_2+R_3 \rightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 4 & -2 & 1 & 0 \\ 0 & 0 & -1 & 1 & -2 & 1 \end{array} \right] \\ &\xrightarrow{-R_3 \leftrightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 4 & -2 & 1 & 0 \\ 0 & 0 & 1 & -1 & 2 & -1 \end{array} \right] \\ &\xrightarrow{\substack{-4R_3+R_2 \rightarrow R_2 \\ 2R_3+R_1 \rightarrow R_1}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 4 & -2 \\ 0 & 1 & 0 & 2 & -7 & 4 \\ 0 & 0 & 1 & -1 & 2 & -1 \end{array} \right] \\ \therefore A^{-1} &= \begin{bmatrix} -1 & 4 & -2 \\ 2 & -7 & 4 \\ -1 & 2 & -1 \end{bmatrix} \end{aligned}$$

Verification

$$\begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 4 & -2 \\ 2 & -7 & 4 \\ -1 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 4 & -2 \\ -2 & -7 & 4 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}.$$

Application

Example. Solve the following system

$$x_1 - 2x_3 = 3$$

$$2x_1 + x_2 = 1$$

$$3x_1 + 2x_2 + x_3 = -2$$

Solution. Let us put the system into the following form $A\mathbf{x} = \mathbf{b}$.

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}.$$

Since A is invertible, the given system has a unique solution

$$\mathbf{x} = A^{-1}\mathbf{b}$$

From the previous example, we have

$$A^{-1} = \begin{bmatrix} -1 & 4 & -2 \\ 2 & -7 & 4 \\ -1 & 2 & -1 \end{bmatrix}$$

So,

$$\mathbf{x} = \begin{bmatrix} -1 & 4 & -2 \\ 2 & -7 & 4 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -9 \\ 1 \end{bmatrix}.$$